NAME: **Edenilson (Eddie) Lopes De Meira**
LANGUAGE: **English**
Clearance: **Secret**

SUMMARY OF QUALIFICATIONS

Highly experienced software engineer and developer with a strong foundation in software development, automation, and technical problem-solving. Proficient in C#, PowerShell scripting, Agile methodologies, and technical documentation. Published author of "How to C# in a Fuzzy World." Adept at working collaboratively in team environments and delivering innovative solutions within tight deadlines. Certified in multiple relevant fields, including PMP, CMSP, and SAFe.

- Programming Languages: C, C++, C#, Python and Java
- Scripting: PowerShell, Ansible
- Development tools: Visual Studio, IntelliJ IDEA, PyCharm, VS Code
- Source Control: Git, GitLab, SVN
- Game Engines: Unity 3D
- Testing and Automation: Regression testing, testing plans, testing cases
- Deployment and Configuration: SSCM, VMware setup
- Technical Documentation: Requirement Specification, Functional Description, Architecture Design, Testing plan, technical manuals, technical investigation and research
- Problem Solving: Root cause analysis, debugging complex systems
- Collaboration and Communication: Excellent interpersonal skills; ability to effectively present technical concepts to non-technical stakeholders.
  Agile Methodologies: Experience working in Agile and SAFe environments; familiarity with Scrum and Kanban practices.

CERTIFICATIONS

- PMP – PMI (09/2024 – 09/2027)
  Project Management Professional – Issued by The Project Management Institute
- CMSP - NTSA (04/2022 – 04/2030)
  Certified Modeling and Simulation Professional – Master Level – Issued by The National Training and Simulation Association
- ITCP – CIPS (01/2026 – Present)
  Information Technology Certified Professional – Issued by Canada's Association of IT Professionals
- ISP – CIPS (09/2021 – Present)
  Information Systems Professional – Issued by Canada's Association of IT Professionals
- SP – Scaled Agile (10/2023 – Present)
  Certified SAFe 6 Practitioner – Issued by Scaled Agile Inc.
- CTech – OACETT (08/2014 – 12/2021)
  Certified Technician – Issued by The Ontario Association of Engineering Technicians and Technologists.

EDUCATION

- Master of Science in Computing and Information Systems
- Athabasca University *09/2021 – 09/2026*

LANGUAGES

- English - Fluent
- Portuguese - Fluent

PROFESSIONAL EXPERIENCE

Senior Software Engineer
ADGA Group Consultants
05/2022 – 06/2025
SIMTIES
Kingston/ON, Canada

- Developed C# applications and 3D simulations using Unity3D and Visual Studio 2022.
  - Develop C# scripts for object models to control all aspects of simulation in terms of animation, physics and 3D modelling components
    - Rigidbody
    - Object colliders (box, sphere, capsule, mesh)
    - Character rigging,
    - UI development,
    - Materials and shaders
- Used HDRP and URP pipelines for development
- Created scriptable objects
- Used Unity's Artificial Intelligence (Navmesh) for character animation
- Use Visual Studio 2022 to work on scripts by producing C# classes, instantiate objects, create functions and properties used by the application.
  - Create WinForms applications
  - WPF (Windows Platform Foundation)
  - Console applications (usually for backend)
- Use plugins such as Obi for cable creation, UMotion-Pro to create animations, Rigify to Unity, and Ardity for embedded systems using Arduino.
- Used Blender 4.0 for 3D modelling (Hard Surface)
  - Work with primitive objects to create more complex models
  - Edit Meshes
    - Vertex, Edge, Face selection
    - Use subdivision and smoothing
    - Created bevels on edges
    - Worked with topology – quads and Ngons
- Utilized Agile for project management and collaboration.
  - Participated and conducted standup meetings
  - Used scrums to approach development
  - Worked with Kanban
  - Worked with Extreme Programming
  - Prepared Product Backlog
  - Created and used burndown/burnup charts in GitLab
  - Participated in sprint reviews
  - Participated in sprint retrospectives
  - Developed Continuous Integration & Continuous Deployment (CI/CD)
- Worked with Scalable Agile Framework (SAFe 6.0).
  - Agile Release Train (ART)
  - Program Increments (PIs)
  - Produced System's Demos
  - Managed Work Backlog
  - Acted as a Product Owner
  - Acted as Scrum Master
- Created and maintained technical documentation, emphasizing test artifacts like test plans and cases.
  - Created and Maintained
    - White papers
    - Technical Manuals
    - Software Architecture Document
    - Software Design Document
    - Software Functional Description Document
    - Software Testing Procedures
    - Factory Acceptance Test Document
    - Software Requirement Specification
- Collaborated with cross-functional teams to troubleshoot issues and define project requirements.
- Trained junior developers and managed project development through its Software Life Development Cycle (SLDC)
  - Planning
  - Analysis

- Development
- Testing
- Deployment
- Maintenance

Program Analyst
ADGA Group Consultants
08/2021 – 05/2022
MODSIM
Kingston/ON, Canada

- Modeling and Simulation;
  - 3D Model Development: Created and optimized 3D models for game and simulation objects.
  - Integration & Interoperability: Integrated objects with various development platforms, ensuring seamless interoperability.
  - Constructive Simulation: Designed and implemented constructive simulation models for complex scenarios.
  - System I/O Programming: Developed input/output components for simulation systems, ensuring efficient data processing and communication.
- Use Java/JavaFx for simulation development;
  - Designed and implemented simulation applications using Java and JavaFX for interactive and real-time visualization.
  - Utilized the Maven framework to manage dependencies and build automation.
  - Automated Windows-based development tasks using PowerShell scripting.
  - Developed Linux-based components and automation scripts using Bash scripting.
  - Managed and configured XML-based framework components to define simulation parameters and system configurations.
- Work in Linux environment, use bash commands;
- Setup Virtual Machines with VMWare;
- Use Git/GitLab for source control and Git Kraken as git client;
- Use Visual Studio, IntelliJ IDEA IDEs and Visual Code for scripting.

SCADA Analyst
Bombardier Transportation
07/2011 – 05/2020
SCADA Development
Kingston/ON, Canada

- Program APIs using C# and Visual Studio;
- Developed backend applications in C# for console applications
- Developed WinForms for custom APIs
- Developed WPF applications for TMS (Train Monitoring Systems)
- Developed interfaces for SCADA systems using ICONICS platform and Genesis64
- Worked with OPC (Object Linking for Process Control) server
  - Used OPCUA and OPCDA for signal iterfaces
- Worked with Microsoft server 2012 setup
- Setup KepWare server for signal control
  - Developed custom interfaces with C# to exchange/control signals between KepWare server and ICONICS platform
- Developed a proprietary application for operation programming at Station control centres
  - Electrical distribution systems
    - Power distribution houses
    - Station lighting system
    - Station security and monitoring control system
  - Auxiliary systems
    - HVAC systems,
    - Exit doors,
    - CCTVs
  - Flow passenger systems;
    - PI/PA systems
    - Billboards
    - Gates

        - Station safety doors for boarding the trains
        - Panel banners

- Produced Documentation:
    - Software Requirements Specification
    - Software Functional Description
    - Software Architectural Design
    - Software Design Document
    - Software Quality Control
    - Factory Acceptance Test
    - Software Testing Procedures
    - Software Testing Reports
    - Request for Information Documents
    - Request for Proposals (for contractors)
    - Business Engineering Design (for internal design and intellectual property)

- Provide technical training for Brazilian client in Portuguese;

- Liasson with the client for changes in customer and engineering requirements;

- Use hybrid project management methodology (Waterfall and Agile) to gather requirements and produce deliverables;

- Design, program and implement code for complex simulation and applications within SCADA.
    - Developed code and program design for SIMEA (Simulation of Electrical and Auxiliary Systems)
    - Developed code, structure and design for DDAS (Data Distributed and Acquisition System – Similar to SCADA but specific for train communication).
    - Developed Data Dictionary for TMS
    - Developed custom applications for train control systems and data acquisition using C# console applications and WPF.

Site Manager
Transformix Engineering
04/2010 – 07/2011
Project Management
Kingston/ON, Canada

- Project Management and Leadership
    - Directed and managed an international engineering team traveling from Canada to Brazil, ensuring smooth operations, resource allocation, and cross-border collaboration.
    - Developed and maintained project schedules using Agile and traditional methodologies, ensuring timely completion of milestones and deliverables.
    - Managed project budgets, tracking expenses, optimizing resource allocation, and ensuring cost-effective execution while maintaining high-quality standards.

- Client and Stakeholder Engagement
    - Acted as the primary liaison with clients, leveraging Agile methodologies to facilitate collaboration, gather requirements, and ensure continuous stakeholder engagement.
    - Conducted regular project updates, sprint reviews, and retrospectives to maintain transparency and align deliverables with client expectations.
    - Represented the company in professional interviews, conferences, and stakeholder meetings, enhancing corporate visibility and strengthening industry relationships.

- Regulatory Compliance and Risk Management
    - Ensured compliance with local Brazilian regulations, coordinating with government agencies to validate engineers' documentation, work permits, and visas.
    - Navigated international regulatory environments, ensuring adherence to industry standards, legal requirements, and safety protocols.
    - Identified and mitigated project risks, proactively addressing compliance, travel, and operational challenges in an international business environment.

- Cross-Cultural Team Management and Training
    - Led technical training and cross-referencing of Brazilian and Canadian engineering standards/norms, ensuring alignment of methodologies and regulatory compliance.
    - Facilitated knowledge transfer sessions, bridging gaps between Canadian and Brazilian engineering teams to promote efficiency and technical consistency.
    - Mentored and supported engineers and project teams, fostering a collaborative and high-performance work environment.

- Represented the company in interviews and conferences.

Instrumentation Technician
Research In Motion - RIM
09/2009 – 04/2010
Instrumentation Maintenance and Troubleshooting

Waterloo/ON, Canada

- Electronic Instrumentation and Calibration Management
  - Managed the calibration of electronic instrumentation, ensuring precision and compliance with industry standards and regulatory requirements.
  - Maintained and updated electronic databases for calibration control, tracking instrumentation performance, calibration schedules, and historical data for audits.
  - Implemented quality control measures, ensuring instrumentation accuracy and minimizing downtime in critical operations.
  - Developed calibration testing procedures and reports.
  - Followed ISO 9000 quality standards for metrology instrumentation and control

- Embedded Systems Development and Debugging
  - Debugged and modified C code for embedded systems, enhancing system performance, reliability, and real-time processing capabilities.
  - Developed and optimized firmware for embedded controllers, ensuring seamless integration with industrial automation systems.
  - Conducted root cause analysis on embedded systems issues, identifying and resolving hardware/software discrepancies.

- Technical Documentation and Compliance
  - Authored and maintained technical reports, procedures, and testing documentation, ensuring clarity and compliance with industry and regulatory standards.
  - Developed and executed test protocols, validating system performance, reliability, and compliance with engineering specifications.
  - Collaborated with cross-functional teams to document system designs, troubleshooting guides, and operational manuals.

- Industrial Automation and PLC Programming
  - Programmed, troubleshot, and optimized Allen-Bradley PLCs in production lines, ensuring efficient automation and minimal downtime.
  - Configured and maintained PLC-controlled systems, enhancing manufacturing process efficiency and reliability.
  - Performed real-time diagnostics on industrial automation systems, ensuring continuous operation and prompt resolution of control system failures.

Other Companies

- Field Technician
  05/2009 – 09/2009 *The Pressure Pipe Inspection Company Mississauga/ON*

- Robotics Technician and Automation Department Lead
  07/2006 – 08/2008 *Hendrickson Springs Intl.* *Stratford/ON*

- Metrology Instrumentation Technician
  04/2000 – 07/2006 *Schaeffler Canada/FAG Aerospace* *Stratford/ON*

- PLC Programmer
  06/1996– 12/1998 *VLM Industrial Automation* *São Paulo - Brazil*

- Quality Control Auditor
  01/1994– 06/1996 *FAG Bearings* *São Paulo - Brazil*

- Metrology Technician
  01/1990– 01/1994 *FAG Bearings* *São Paulo – Brazil*

- Industrial Electrician
  08/1984– 01/1990 *FAG Bearings* *São Paulo - Brazil*

PUBLICATIONS

- *Book:* How to C# in a Fuzzy World
  July/2023 Kindle Direct Publishing
  This book is a practical guide for navigating the uncertainties of real-world C# development. It covers key concepts like OOP, data structures, and debugging while focusing on problem-solving in ambiguous situations. The book emphasizes hands-on learning, clean code practices, and adaptability, offering a clear approach to mastering C# in a changing tech environment.

- *Thesis:* Simulation Interoperability using AI-Driven Frameworks.
  In-Progress
  Athabasca University – School of Computing and Information Systems - Focus: This paper emphasizes how AI can be used to enhance the interoperability of different simulation systems. It focuses more on how various simulation platforms can work together seamlessly, even if they come from different developers or use different architectures.